



AIRLINE DATABASE

PROJECT NAME:

“Airline Database Management System Development”

TEAM MEMBERS:

Ahmed Mostafa Hussein

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1-AIRPORT

- **ATTRIBUTES:** AIRPORTCODE (PRIMARY KEY), AIRPORTNAME, CITY, STATE
- **RELATIONSHIPS:**
 - FLIGHT_LEG: ONE AIRPORT CAN HAVE MULTIPLE DEPARTURE FLIGHT_LEG (ONE-TO-MANY)
 - FLIGHT_LEG: ONE AIRPORT CAN HAVE MULTIPLE ARRIVAL FLIGHT_LEG (ONE-TO-MANY)
 - AIRPLANE_TYPE: MULTIPLE AIRPORT CAN HAVE MULTIPLE AIRPLANE_TYPES IT CAN LAND (MANY-TO-MANY)

2-FLIGHT

- **ATTRIBUTES:** FLIGHTNUMBER (PRIMARY KEY), AIRLINE, WEEKDAYS
- **RELATIONSHIPS:**
 - FLIGHT_LEG: ONE FLIGHT CAN HAVE MULTIPLE FLIGHT_LEG (ONE-TO-MANY)

3-FLIGHTLEG

- **ATTRIBUTES:** LEG_NUMBER (WEAK KEY), DEPARTUREAIRPORT, SCHEDULEDDEPARTURETIME, ARRIVALAIRPORT, SCHEDULEDARRIVALTIME
- **RELATIONSHIPS:**
 - FLIGHT: MULTIPLE FLIGHT_LEG IS ASSOCIATED WITH ONE FLIGHT (MANY-TO-ONE)
 - LEG_INSTANCE: ONE FLIGHT_LEG CAN HAVE MULTIPLE LEG_INSTANCE (ONE-TO-MANY)
 - DEPARTURE_AIRPORT: MULTIPLE FLIGHT_LEG IS ASSOCIATED WITH ONE DEPARTURE_AIRPORT (MANY-TO-ONE)
 - ARRIVAL_AIRPORT: MULTIPLE FLIGHT_LEG IS ASSOCIATED WITH ONE ARRIVAL_AIRPORT (MANY-TO-ONE)

4-LEGINSTANCE

- **ATTRIBUTES:** DATE (WEAK KEY), ACTUALDEPARTURETIME, ACTUALARRIVALTIME, NUMBEROFAVAILABLESEATS
- **RELATIONSHIPS:**
 - FLIGHT_LEG: MULTIPLE LEG_INSTANCE IS ASSOCIATED WITH ONE FLIGHT_LEG (MANY-TO-ONE)
 - RESERVATION: ONE LEG_INSTANCE CAN HAVE MULTIPLE RESERVATIONS (ONE-TO-MANY)
 - AIRPLANE: MULTIPLE LEG_INSTANCE IS ASSOCIATED WITH ONE AIRPLANE (MANY-TO-ONE)

5-RESERVATION

- **ATTRIBUTES:** CUSTOMERNAME, CUSTOMERPHONE, SEATNUMBER (WEAK KEY)
- **RELATIONSHIPS:**
 - LEG_INSTANCE: MULTIPLE RESERVATION IS ASSOCIATED WITH ONE LEG_INSTANCE (MANY-TO-ONE)

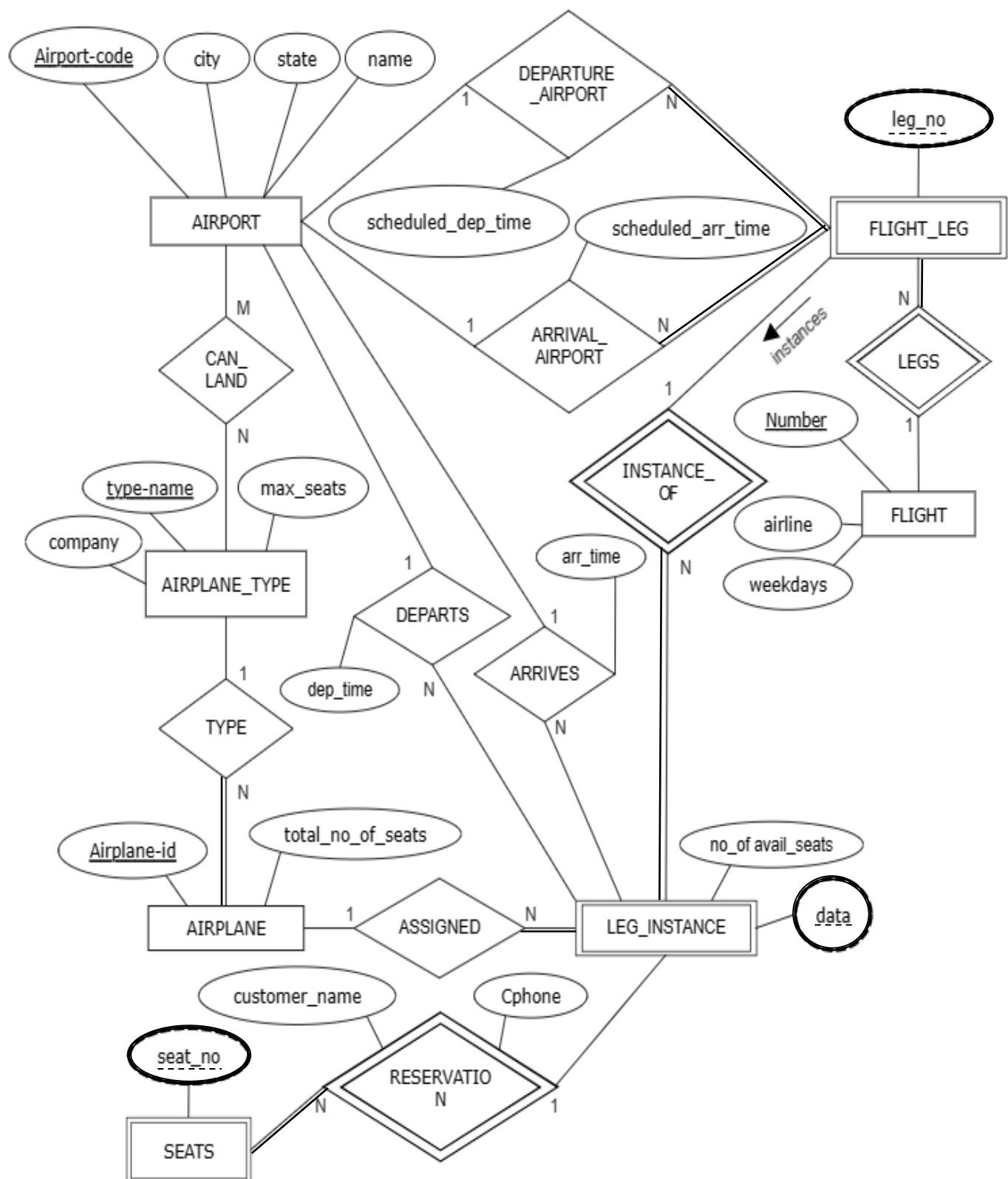
6-AIRPLANE

- **ATTRIBUTES:** AIRPLANEID (PRIMARY KEY), TOTALNUMBEROFSEATS
- **RELATIONSHIPS:**
 - LEG_INSTANCE: ONE AIRPLANE CAN HAVE MULTIPLE LEG_INSTANCES (ONE-TO-MANY)
 - AIRPLANE_TYPE: MULTIPLE AIRPLANE IS ASSOCIATED WITH ONE AIRPLANE_TYPE (MANY-TO-ONE)

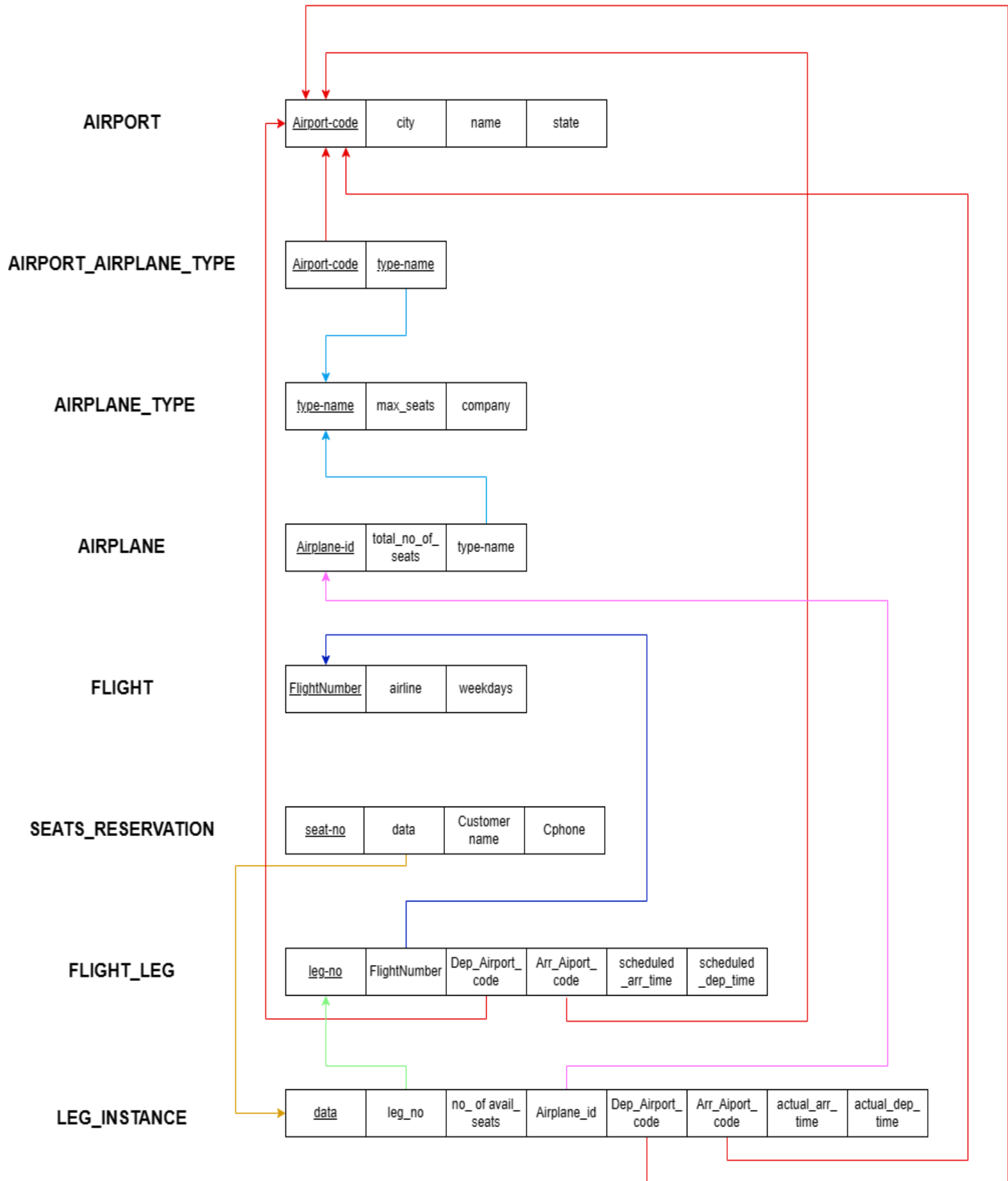
7-AIRPLANETYPE

- **ATTRIBUTES:** TYPENAME (PRIMARY KEY), MANUFACTURINGCOMPANY, MAXIMUMNUMBEROFSEATS
- **RELATIONSHIPS:**
 - AIRPLANE: ONE AIRPLANE_TYPE CAN HAVE MULTIPLE AIRPLANES (ONE-TO-MANY)
 - AIRPORT: MULTIPLE AIRPLANE_TYPE CAN LAND AT MULTIPLE AIRPORTS (MANY-TO-MANY)

1-THE ENTITY RELATIONSHIP (ER) MODEL

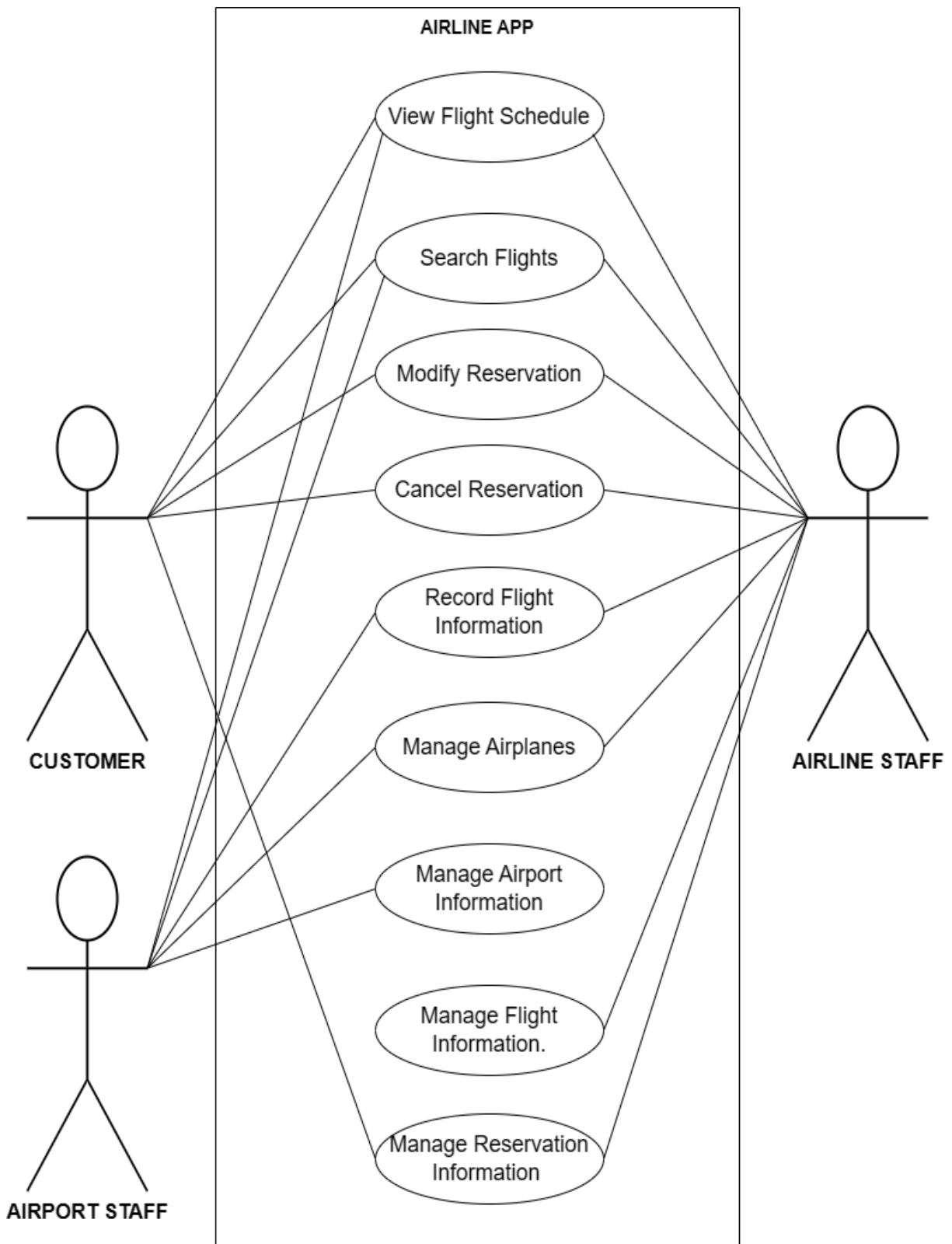


2-THE REFERENTIAL CONSTRAINTS SCHEMA



3-UML DIAGRAMS

• THE USE CASE DIAGRAM



- **VIEW FLIGHT SCHEDULE:** THIS USE CASE WOULD ALLOW ALL ACTORS (CUSTOMER, AIRLINE STAFF, AIRPORT STAFF) TO VIEW THE FLIGHT SCHEDULE FOR A SPECIFIC AIRLINE, INCLUDING THE **FLIGHT NUMBERS, DEPARTURE AND ARRIVAL AIRPORTS, SCHEDULED DEPARTURE AND ARRIVAL TIMES, AND DAYS OF THE WEEK** THE FLIGHT OPERATES.
- **SEARCH FLIGHTS:** THIS USE CASE WOULD ALLOW ALL ACTORS (CUSTOMER, AIRLINE STAFF, AIRPORT STAFF) TO SEARCH FOR FLIGHTS BASED ON VARIOUS CRITERIA, SUCH AS **DEPARTURE AND ARRIVAL AIRPORTS, DATES, AND NUMBER OF AVAILABLE SEATS.**
- **MODIFY RESERVATION:** THIS USE CASE WOULD ALLOW (CUSTOMER, AIRLINE STAFF) TO MODIFY THEIR EXISTING RESERVATIONS, SUCH AS **CHANGING FLIGHT DATES** OR SELECTING **DIFFERENT SEATS.**
- **CANCEL RESERVATION:** THIS USE CASE WOULD ALLOW (CUSTOMER, AIRLINE STAFF) TO CANCEL THEIR EXISTING RESERVATIONS.
- **RECORD FLIGHT INFORMATION:** THIS USE CASE WOULD ALLOW (AIRLINE STAFF, AIRPORT STAFF) TO RECORD **ACTUAL DEPARTURE AND ARRIVAL TIMES,** AS WELL AS ANY OTHER NECESSARY INFORMATION, FOR EACH **FLIGHT LEG INSTANCE.**
- **MANAGE AIRPLANES:** THIS USE CASE WOULD ALLOW (AIRLINE STAFF, AIRPORT STAFF) TO MANAGE INFORMATION ABOUT AIRPLANES, INCLUDING THE **AIRPLANE ID, TYPE, AND NUMBER OF SEATS.**
- **MANAGE AIRPORT INFORMATION:** THIS USE CASE WOULD ALLOW AIRPORT STAFF TO MANAGE INFORMATION ABOUT AIRPORTS, INCLUDING THE **AIRPORT CODE, NAME, CITY, AND STATE.**
- **MANAGE FLIGHT INFORMATION:** THIS USE CASE WOULD ALLOW AIRLINE STAFF TO MANAGE INFORMATION ABOUT FLIGHTS, INCLUDING THE **FLIGHT NUMBER, AIRLINE, AND DAYS OF THE WEEK** THE FLIGHT OPERATES.
- **MANAGE RESERVATION INFORMATION:** THIS USE CASE WOULD ALLOW (AIRLINE STAFF, CUSTOMER) TO MANAGE INFORMATION ABOUT CUSTOMER RESERVATIONS, INCLUDING THE **CUSTOMER NAME, PHONE NUMBER, AND SEAT NUMBER(S).**

• USE CASE SCENARIO

1-FIRST CASE

Use Case Name: *Modify Reservation*

Actor(s): *Customer, Airline Staff*

Area: *Reservation*

Main Scenario (steps):

1. *Log in to the system with their username and ID*
2. *Go to reservation modification page*
3. *Select existing reservation*
4. *Modify reservation details (e.g. change the flight, change the date, change the seat, etc.)*
5. *Confirm changes*
6. *System updates reservation in the database*

Preconditions: *A valid reservation exists and has not already been flown*

Postconditions: *The reservation has been modified as requested*

2-SECOND CASE

Use Case Name: *Manage Airport Information*

Actor(s): *Airport Staff*

Area: *Airport*

Main Scenario (steps):

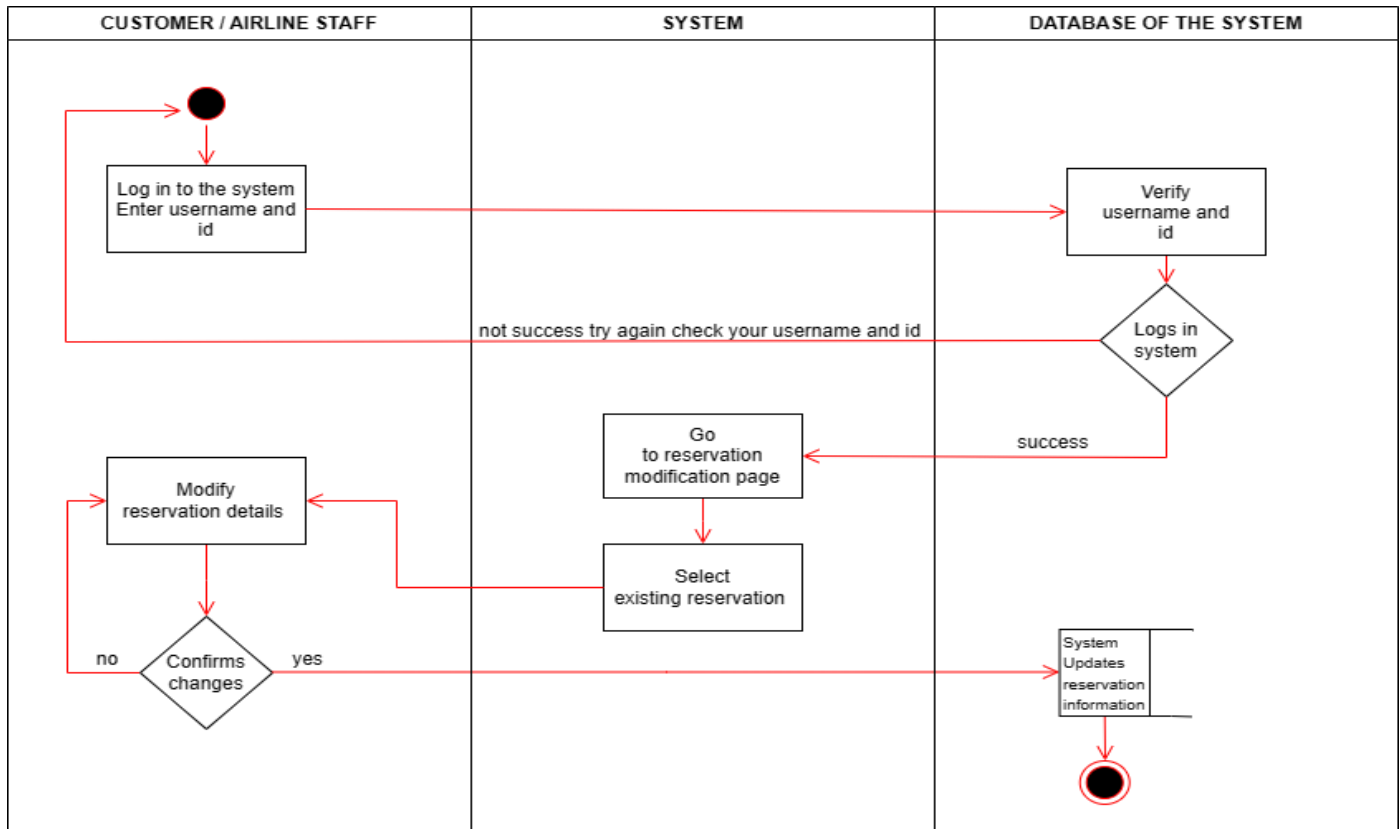
1. *Log in to the system with their username and ID*
2. *Go to airport management page*
3. *Select airport to manage*
4. *Modify airport details (e.g. change the airport name, change the city, change the state, etc.)*
5. *Confirm changes*
6. *System updates airport information in the database*

Preconditions: *Airport Staff has accessed the airline database system*

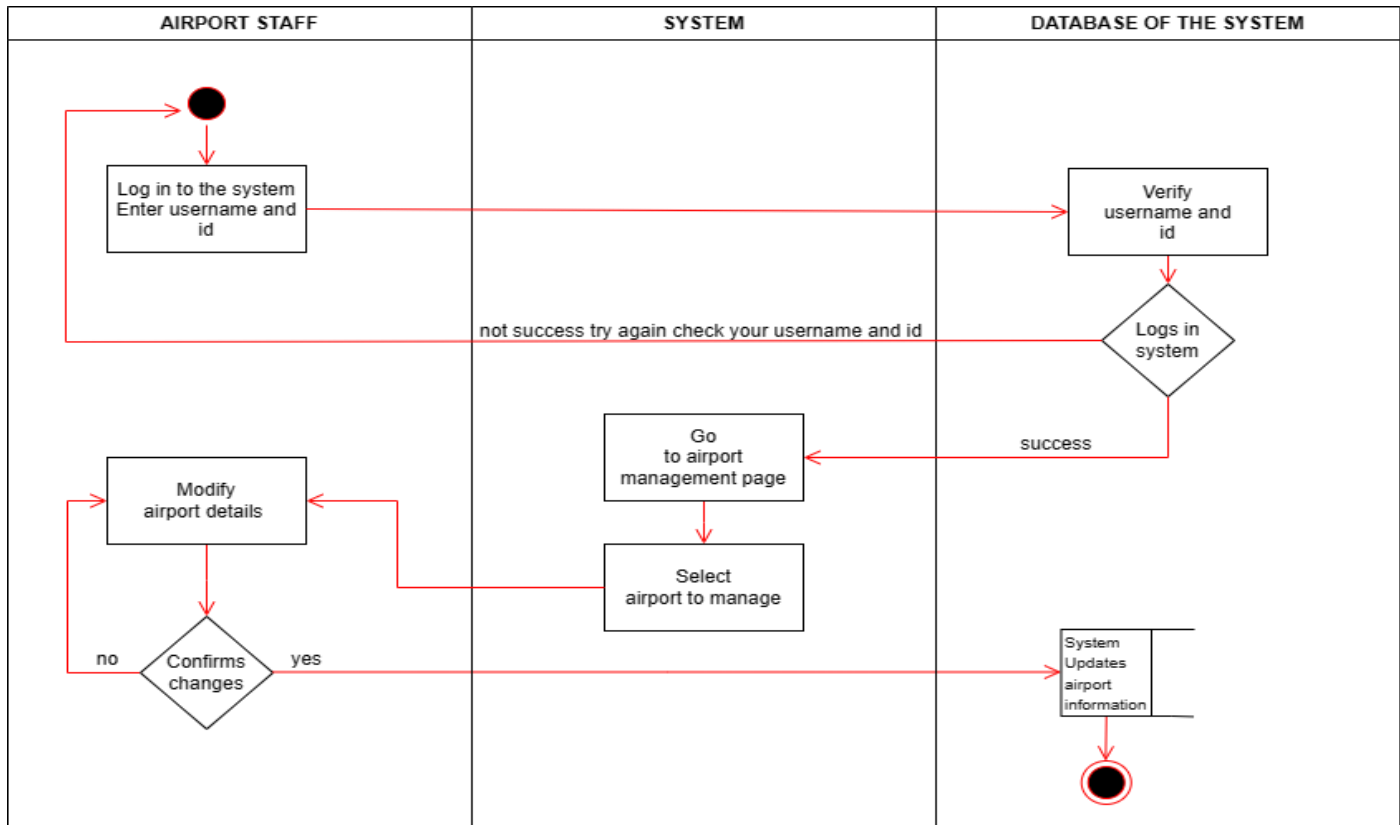
Postconditions: *The airport information has been updated or a new airport added as requested*

• ACTIVITY DIAGRAM

1-FIRST CASE

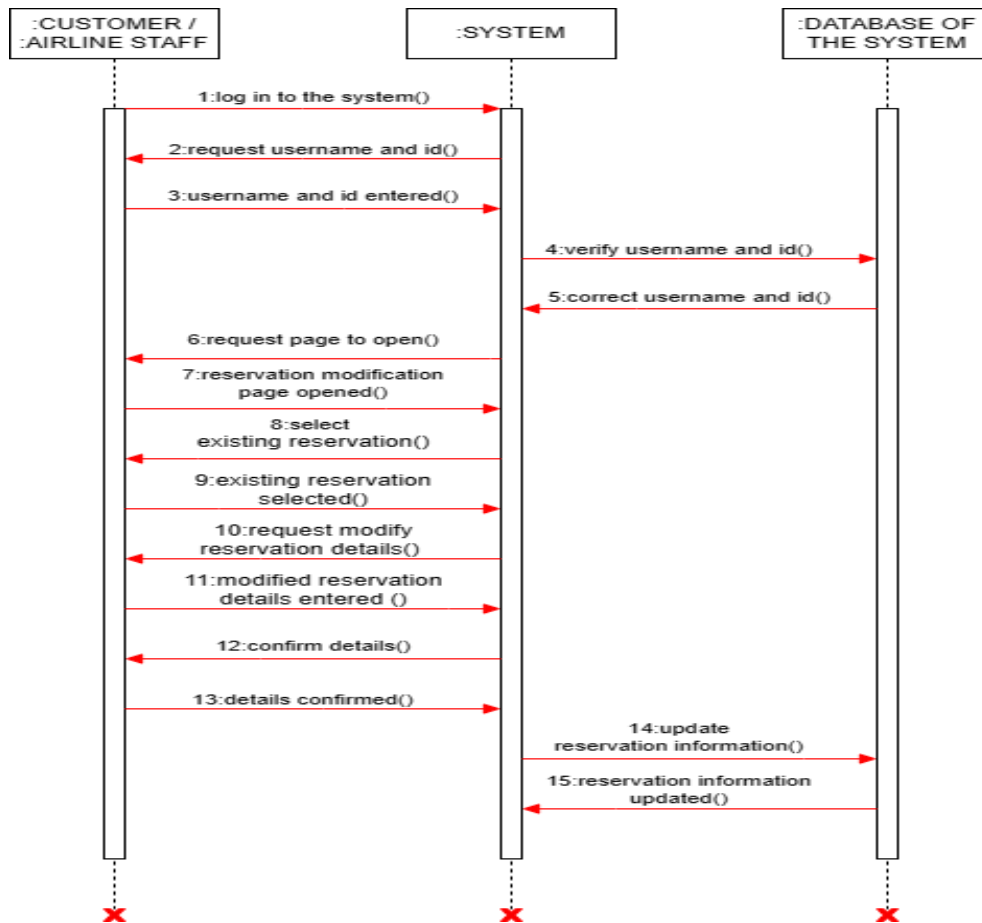


2-SECOND CASE

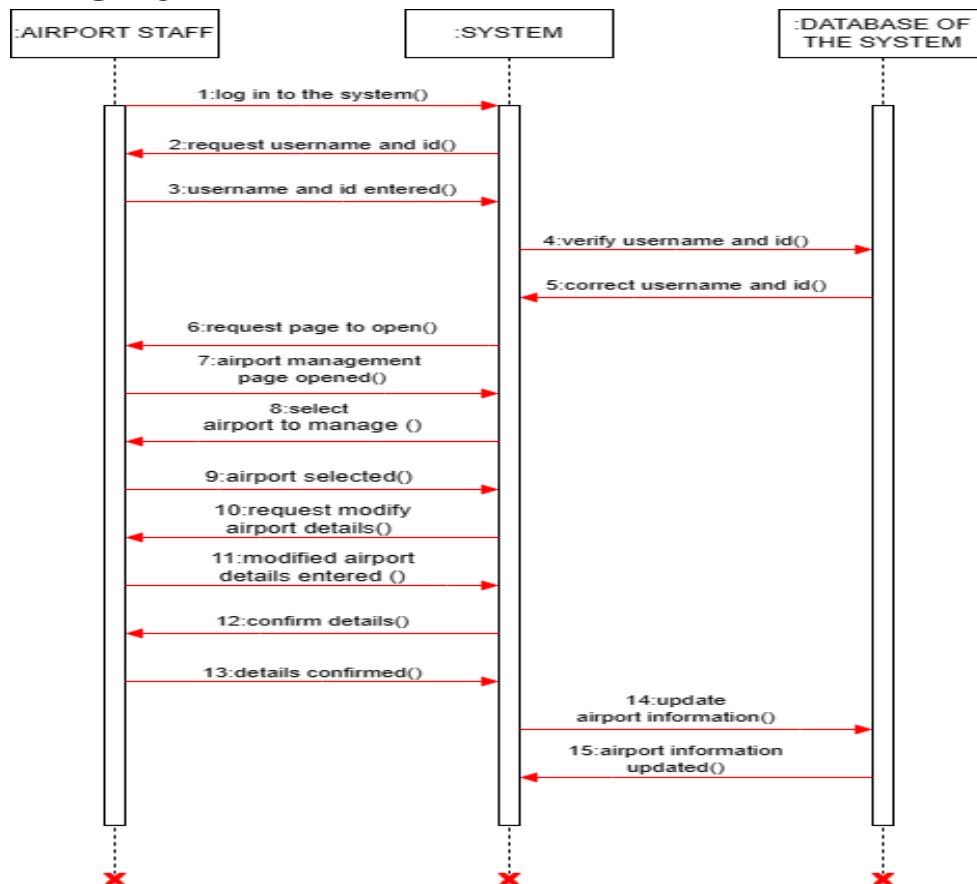


• SEQUENCE DIAGRAM

1-FIRST CASE

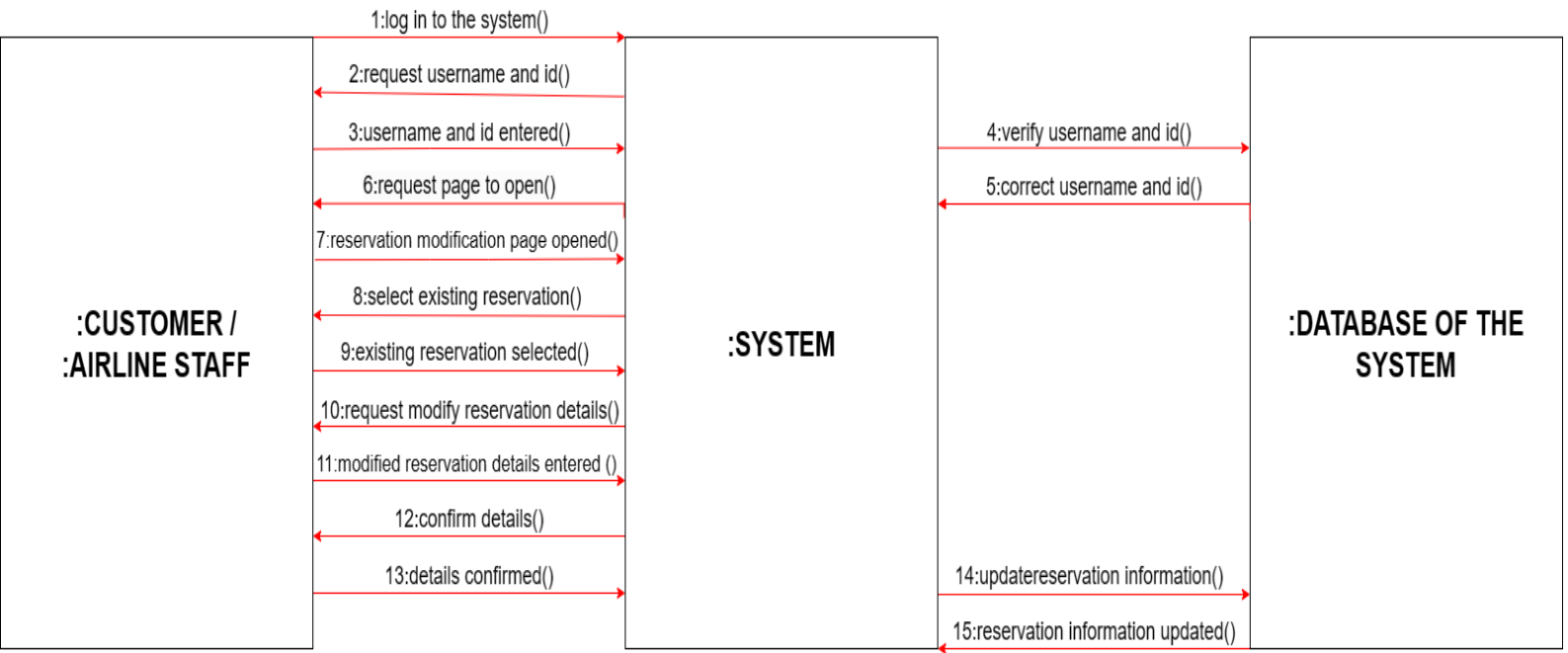


2-SECOND CASE

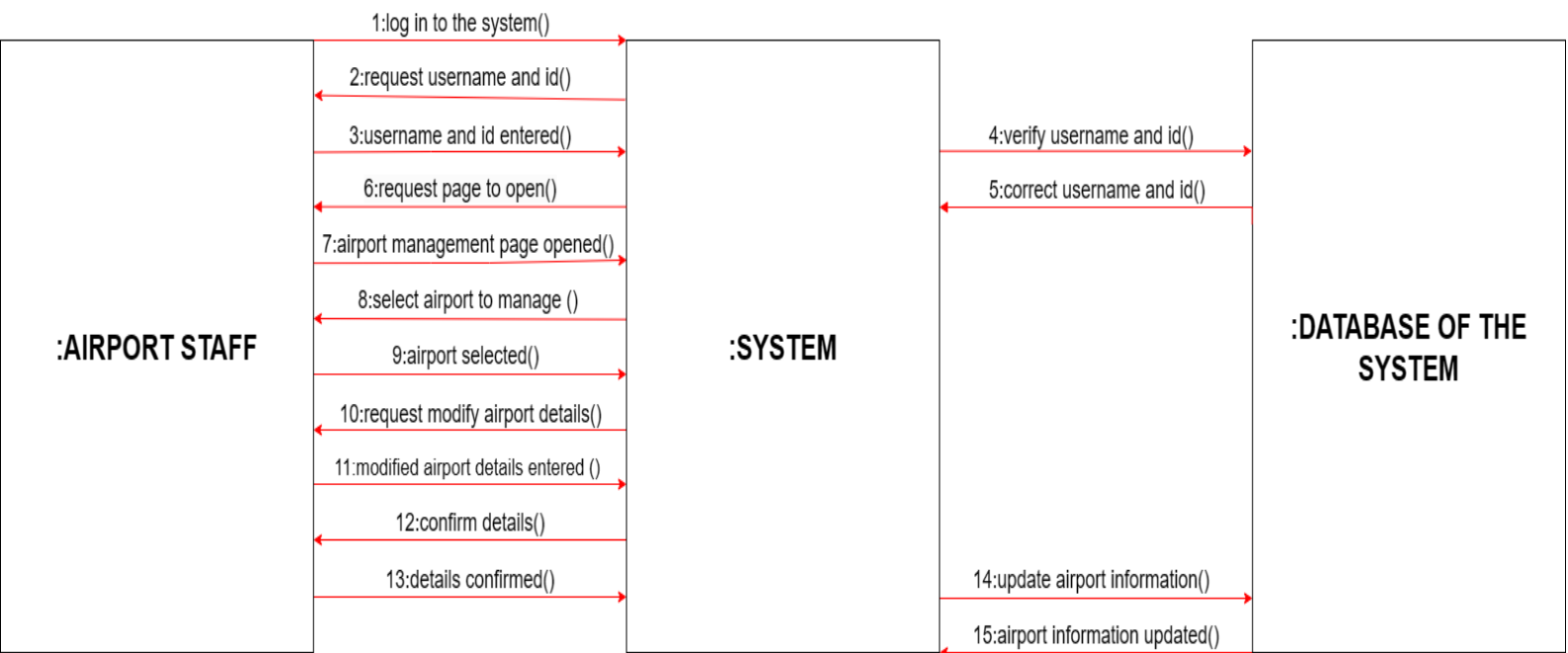


• COMMUNICATION DIAGRAM

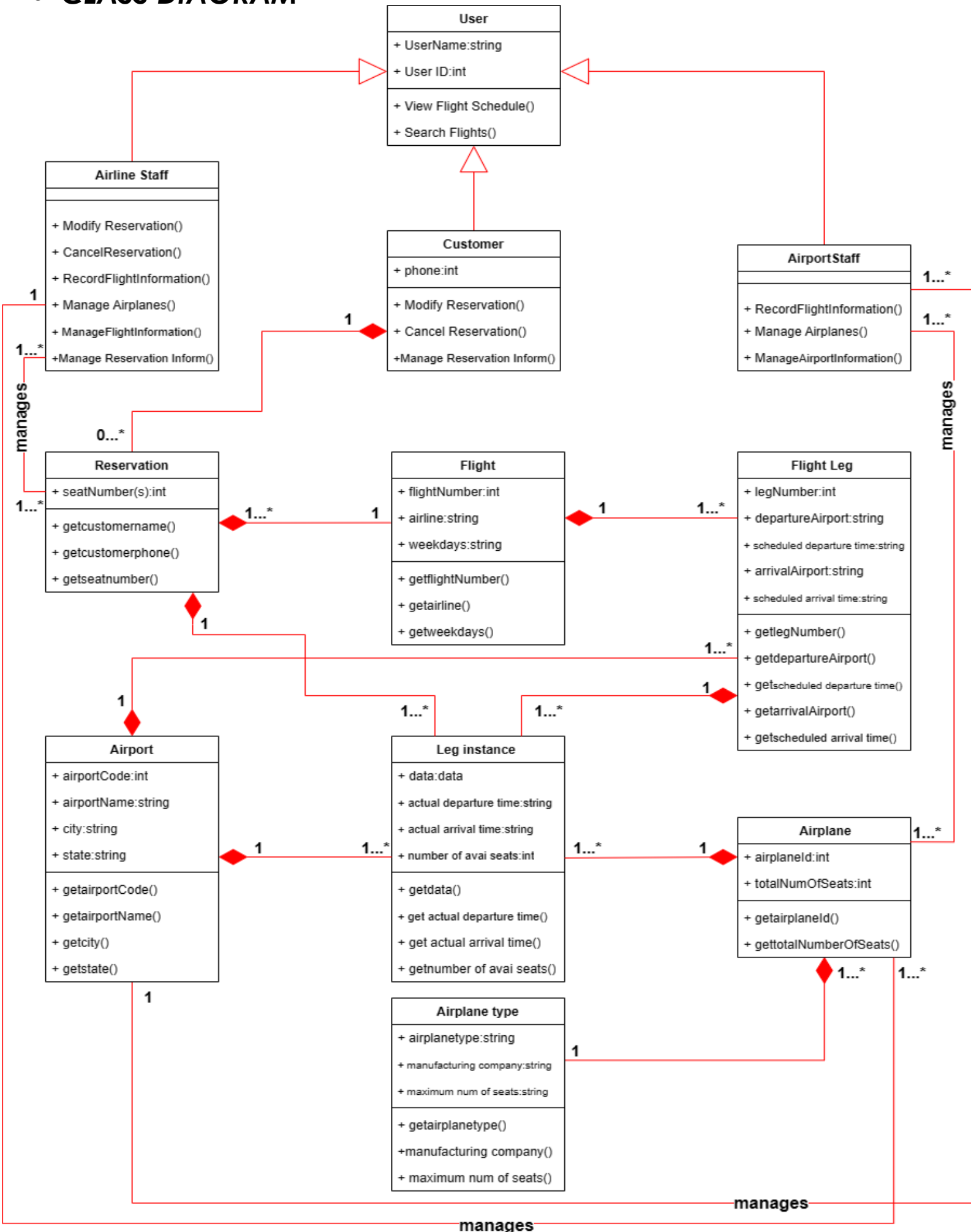
1-FIRST CASE



2-SECOND CASE



• CLASS DIAGRAM



4-USER-FRIENDLY INTERFACE

• FIRST CODING OF SQL

```
SQLQuery1.sql - (local).Airline_Database (AHMED-MUSTAFAS-\\Ahmed mostafa (67)) - Microsoft SQL Server Management Studio
File Edit View Query Project Tools Window Help
Airline_Database Execute
Object Explorer
Connect
(SQL Server 16.0.1050.5 - AHMED-M
Databases
System Databases
Database Snapshots
Airline_Database
Database Diagrams
Tables
System Tables
FileTables
External Tables
Graph Tables
dbo.AIRPLANE
dbo.AIRPLANE_TYPE
dbo.AIRPORT
dbo.AIRPORT_AIRPLANE_
dbo.FLIGHT
dbo.FLIGHT_LEG
dbo.LEG_ISTANCE
dbo.SEATS_RESERVATION
Dropped Ledger Tables
Views
External Resources
Synonyms
Programmability
Query Store
Service Broker
Storage
Security
Server Objects
Replication
Always On High Availability
Management
Integration Services Catalogs
SQL Server Agent / Agent VDs first
SQLQuery1.sql - (\\hmed mostafa (67))
create database Airline_Database
use Airline_Database
create table AIRPORT(
Airportcode int primary key,
Airportname varchar(30) not null,
city varchar(30) not null,
state varchar(30) not null,
);
create table AIRPLANE_TYPE(
type_name varchar(30) primary key,
maximum_number_of_seats int not null,
manufacturingcompany varchar(30) null,
);
create table AIRPORT_AIRPLANE_TYPE(
Airportcode int not null,
type_name varchar(30) not null,
constraint AIRPORT_fk foreign key (Airportcode)
references Airport (Airportcode),
constraint AIRPLANE_TYPE_fk foreign key (type_name)
references AIRPLANE_TYPE (type_name),
);
create table AIRPLANE(
AirplaneId int primary key,
total_no_of_seats int not null,
type_name varchar(30) not null,
constraint AIRPLANE_TYPE_AIRPLANE_fk foreign key (type_name)
references AIRPLANE_TYPE (type_name),
);
create table FLIGHT(
FlightNumber int primary key,
Airline varchar(30) not null,
weekdays varchar(30) not null,
);
119 %
Messages
Query executed successfully.
(local) (16.0 RTM) AHMED-MUSTAFAS-\\Ahmed ... Airline_Database 00:00:00 0 rows
```

```
SQLQuery1.sql - (local).Airline_Database (AHMED-MUSTAFAS-\\Ahmed mostafa (67)) - Microsoft SQL Server Management Studio
File Edit View Query Project Tools Window Help
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(SQL Server 16.0.1050.5 - AHMED-M
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dbo.AIRPORT_AIRPLANE_
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Integration Services Catalogs
SQL Server Agent / Agent VDs first
SQLQuery1.sql - (\\hmed mostafa (67))
create table FLIGHT(
FlightNumber int primary key,
Airline varchar(30) not null,
weekdays varchar(30) not null,
);
create table FLIGHT_LEG(
LegNumber int primary key,
Departure_Airportcode int not null,
Arrival_Airportcode int not null,
FlightNumber int not null,
Scheduled_departure_time time not null,
Scheduled_arrival_time time not null,
constraint AIRPORT_FLIGHT_LEG_Departure_fk foreign key (Departure_Airportcode)
references Airport (Airportcode),
constraint AIRPORT_FLIGHT_LEG_Arrival_fk foreign key (Arrival_Airportcode)
references Airport (Airportcode),
constraint FLIGHT_FLIGHT_LEG_fk foreign key (FlightNumber)
references FLIGHT (FlightNumber),
);
create table LEG_ISTANCE(
Data varchar(30) primary key,
Actual_departure_time time not null,
Actual_arrival_time time not null,
Number_of_available_seats int not null,
LegNumber int not null,
AirplaneId int not null,
Departure_Airportcode int not null,
Arrival_Airportcode int not null,
constraint FLIGHT_LEG_LEG_ISTANCE_fk foreign key (LegNumber)
references FLIGHT_LEG (LegNumber),
);
119 %
Messages
Query executed successfully.
(local) (16.0 RTM) AHMED-MUSTAFAS-\\Ahmed ... Airline_Database 00:00:00 0 rows
```



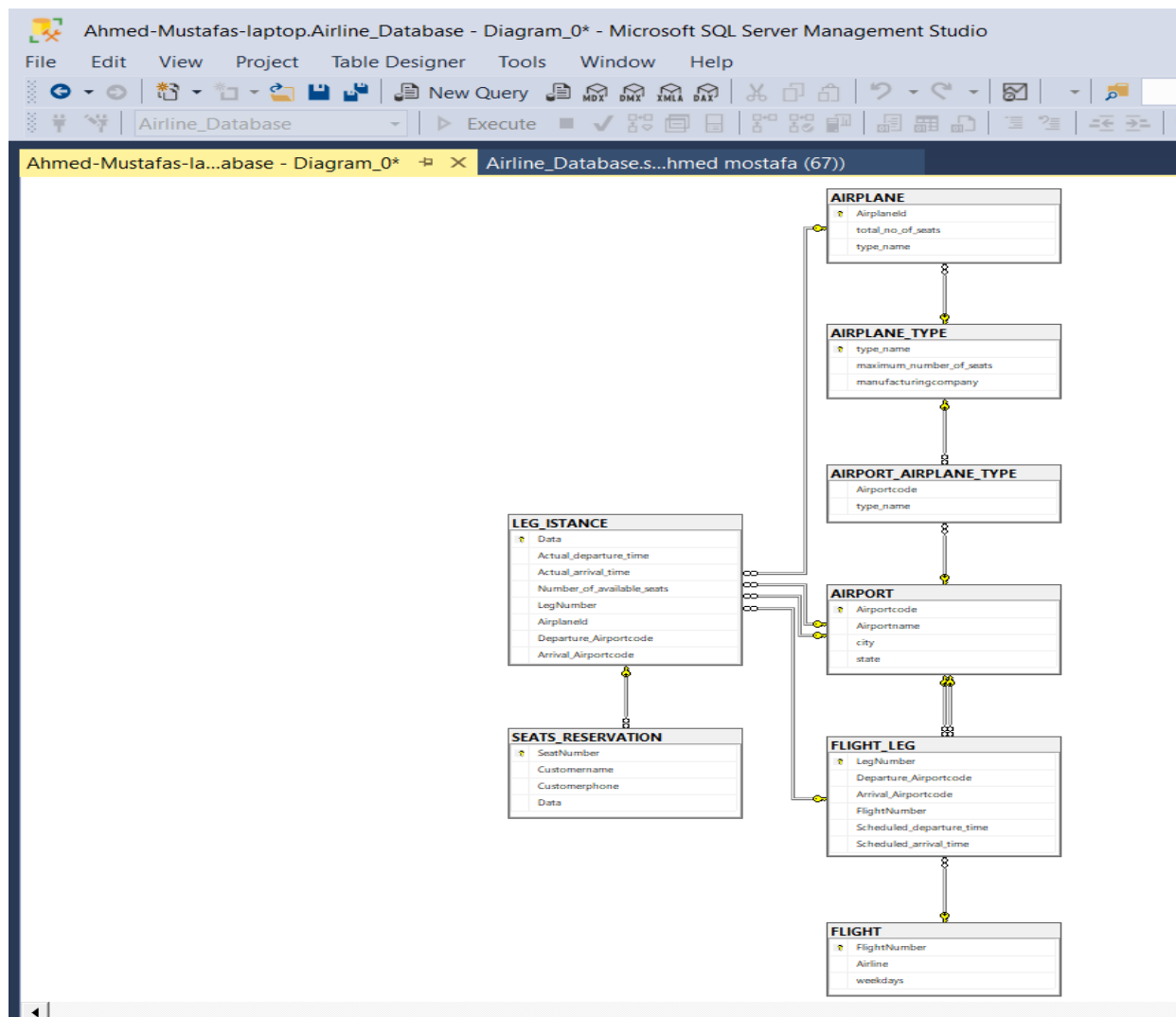
```
SQLQuery1.sql - (local).Airline_Database (AHMED-MUSTAFAS-Ahmed mostafa (67))* - Microsoft SQL Server Management Studio
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New Query Execute
Airline_Database

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SQL Server Agent (Agent YDr dirst)

SQLQuery1.sql - (local).Airline_Database (AHMED-MUSTAFAS-Ahmed mostafa (67))*
);
create table LEG_INSTANCE(
Data varchar(30) primary key,
Actual_departure_time time not null,
Actual_arrival_time time not null,
Number_of_available_seats int not null,
LegNumber int not null,
AirplaneId int not null,
Departure_Airportcode int not null,
Arrival_Airportcode int not null,
constraint FLIGHT_LEG_LEG_INSTANCE_fk foreign key (LegNumber)
references FLIGHT_LEG(LegNumber),
constraint AIRPLANE_LEG_INSTANCE_fk foreign key (AirplaneId)
references AIRPLANE(AirplaneId),
constraint AIRPORT_LEG_INSTANCE_Departure_fk foreign key (Departure_Airportcode)
references Airport (Airportcode),
constraint AIRPORT_LEG_INSTANCE_Arrival_fk foreign key (Arrival_Airportcode)
references Airport (Airportcode),
);
create table SEATS_RESERVATION(
SeatNumber int primary key,
Customername varchar(30)not null,
Customerphone int not null,
Data varchar(30)not null,
constraint LEG_INSTANCE_SEATS_RESERVATION_fk foreign key(Data)
references LEG_INSTANCE (Data),
);

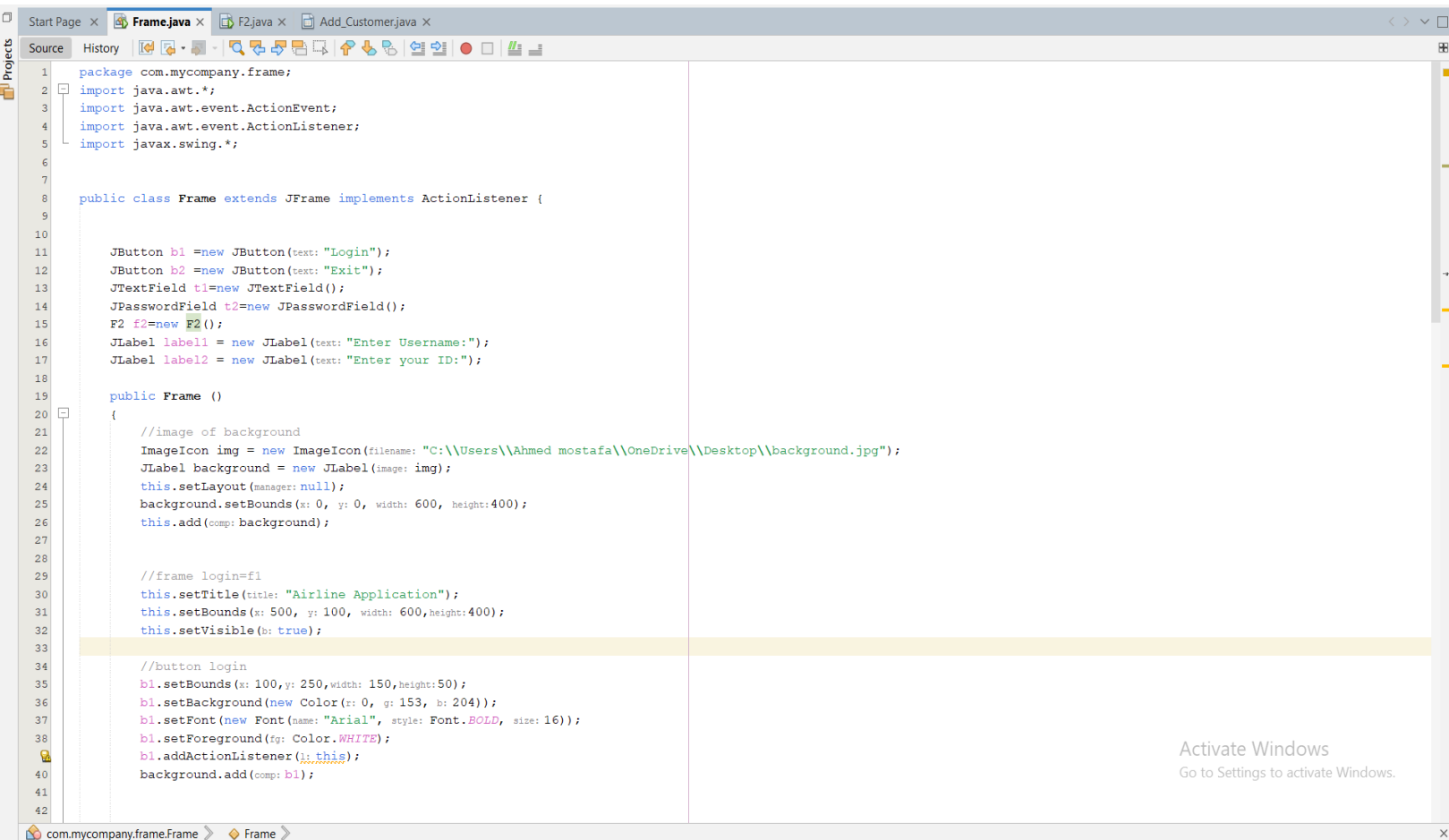
119 %
Messages
Query executed successfully.
(local) (16.0 RTM) AHMED-MUSTAFAS-Ahmed ... Airline_Database 00:00:00 0 rows
```

• DATABASE DIAGRAM

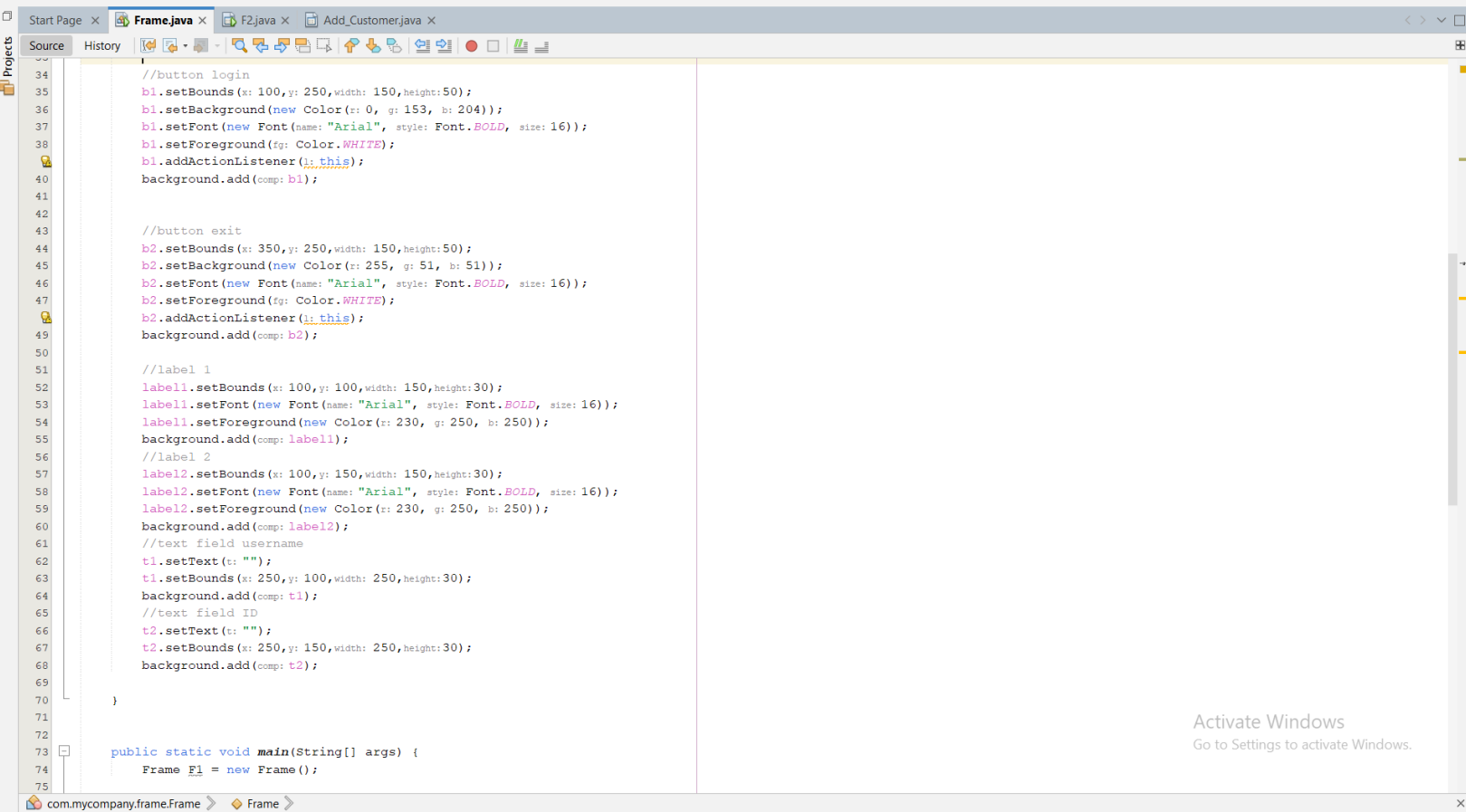


• SECOND CODING OF JAVA

➤ FIRST FRAME WITHOUT DRAG AND DROP



```
1 package com.mycompany.frame;
2 import java.awt.*;
3 import java.awt.event.ActionEvent;
4 import java.awt.event.ActionListener;
5 import javax.swing.*;
6
7
8 public class Frame extends JFrame implements ActionListener {
9
10
11     JButton b1=new JButton(text: "Login");
12     JButton b2=new JButton(text: "Exit");
13     JTextField t1=new JTextField();
14     JPasswordField t2=new JPasswordField();
15     F2 f2=new F2();
16     JLabel label1= new JLabel(text: "Enter Username:");
17     JLabel label2= new JLabel(text: "Enter your ID:");
18
19     public Frame ()
20     {
21         //image of background
22         ImageIcon img = new ImageIcon(filename: "C:\\Users\\Ahmed mostafa\\OneDrive\\Desktop\\background.jpg");
23         JLabel background = new JLabel(image: img);
24         this.setLayout(manager: null);
25         background.setBounds(x: 0, y: 0, width: 600, height:400);
26         this.add(comp: background);
27
28         //frame login=f1
29         this.setTitle(title: "Airline Application");
30         this.setBounds(x: 500, y: 100, width: 600,height:400);
31         this.setVisible(b: true);
32
33
34         //button login
35         b1.setBounds(x: 100,y: 250,width: 150,height:50);
36         b1.setBackground(new Color(r: 0, g: 153, b: 204));
37         b1.setFont(new Font(name: "Arial", style: Font.BOLD, size: 16));
38         b1.setForeground(cg: Color.WHITE);
39         b1.addActionListener(i: this);
40         background.add(comp: b1);
41
42
```



```
34         //button login
35         b1.setBounds(x: 100,y: 250,width: 150,height:50);
36         b1.setBackground(new Color(r: 0, g: 153, b: 204));
37         b1.setFont(new Font(name: "Arial", style: Font.BOLD, size: 16));
38         b1.setForeground(cg: Color.WHITE);
39         b1.addActionListener(i: this);
40         background.add(comp: b1);
41
42
43         //button exit
44         b2.setBounds(x: 350,y: 250,width: 150,height:50);
45         b2.setBackground(new Color(r: 255, g: 51, b: 51));
46         b2.setFont(new Font(name: "Arial", style: Font.BOLD, size: 16));
47         b2.setForeground(cg: Color.WHITE);
48         b2.addActionListener(i: this);
49         background.add(comp: b2);
50
51         //label 1
52         label1.setBounds(x: 100,y: 100,width: 150,height:30);
53         label1.setFont(new Font(name: "Arial", style: Font.BOLD, size: 16));
54         label1.setForeground(new Color(r: 230, g: 250, b: 250));
55         background.add(comp: label1);
56         //label 2
57         label2.setBounds(x: 100,y: 150,width: 150,height:30);
58         label2.setFont(new Font(name: "Arial", style: Font.BOLD, size: 16));
59         label2.setForeground(new Color(r: 230, g: 250, b: 250));
60         background.add(comp: label2);
61         //text field username
62         t1.setText(t: "");
63         t1.setBounds(x: 250,y: 100,width: 250,height:30);
64         background.add(comp: t1);
65         //text field ID
66         t2.setText(t: "");
67         t2.setBounds(x: 250,y: 150,width: 250,height:30);
68         background.add(comp: t2);
69
70     }
71
72
73     public static void main(String[] args) {
74         Frame f1 = new Frame();
75
```

```
Start Page x Frame.java x
Source History
56 //label 1
57 label1.setBounds(x: 100, y: 100, width: 150, height: 30);
58 label1.setFont(new Font(name: "Arial", style: Font.BOLD, size: 16));
59 label1.setForeground(new Color(r: 230, g: 250, b: 250));
60 background.add(comp: label1);
61 //label 2
62 label2.setBounds(x: 100, y: 150, width: 150, height: 30);
63 label2.setFont(new Font(name: "Arial", style: Font.BOLD, size: 16));
64 label2.setForeground(new Color(r: 230, g: 250, b: 250));
65 background.add(comp: label2);
66 //text field username
67 t1.setText(t: "");
68 t1.setBounds(x: 250, y: 100, width: 250, height: 30);
69 background.add(comp: t1);
70 //text field ID
71 t2.setText(t: "");
72 t2.setBounds(x: 250, y: 150, width: 250, height: 30);
73 background.add(comp: t2);
74
75 }
76
77
78 public static void main(String[] args) {
79     Frame f1 = new Frame();
80
81
82 }
83
84
85 @Override
public void actionPerformed(ActionEvent e)
```

Activate Windows
Go to Settings to activate Windows.

com.mycompany.frame.Frame > actionPerformed > if (e.getSource() == b1) else if (e.getSource() == b2) >

Run (Frame) X (14 more...) 107:14 INS Unix (LF)

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help Search (Ctrl+I)

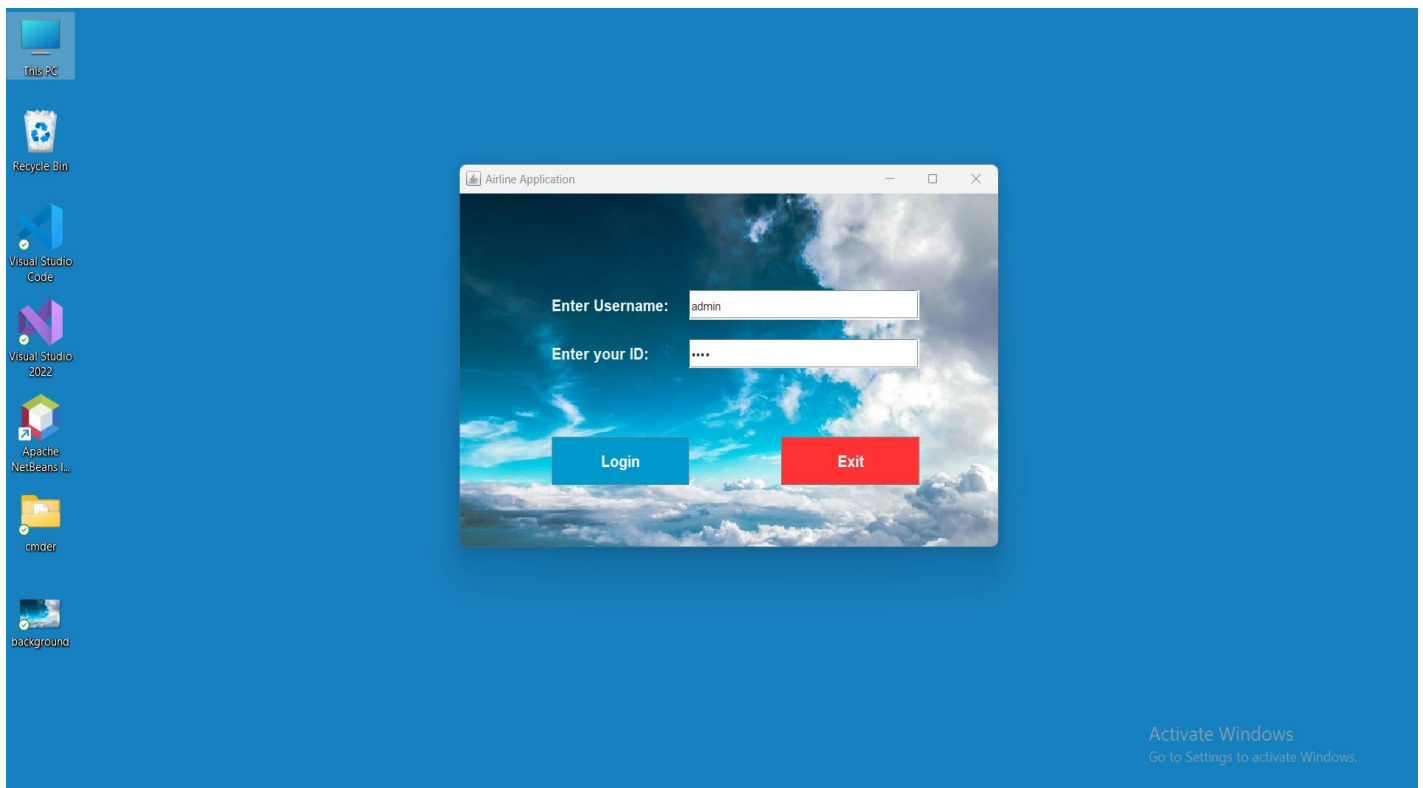
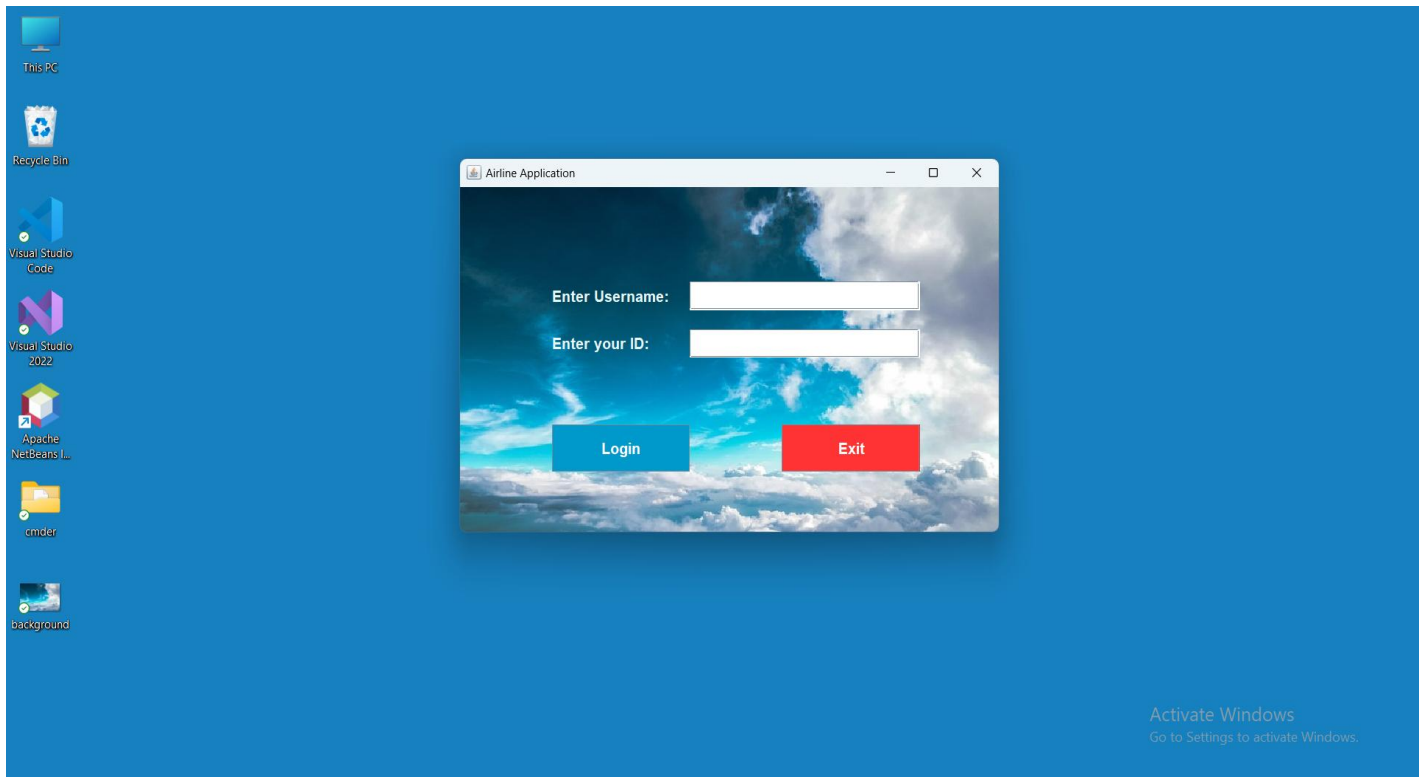
```
Start Page x Frame.java x
Source History
84
85 @Override
86 public void actionPerformed(ActionEvent e)
87 {
88     if (e.getSource() == b1)
89     {
90         // check if username and ID are correct
91         String username = t1.getText();
92         String id = new String(value: t2.getPassword());
93
94         if (username.equals(anObject: "admin") && id.equals(anObject: "1234"))
95         {
96             // display success message and show the new frame
97             JOptionPane.showMessageDialog(parentComponent: this, message: "Login Successful!");
98             f2.setVisible(b: true);
99             this.setVisible(b: false);
100         }
101         else
102         {
103             // display error message
104             JOptionPane.showMessageDialog(parentComponent: this, message: "Username or ID is incorrect. Please try again.");
105         }
106     }
107     else if (e.getSource() == b2)
108     {
109         System.exit(status: 0);
110     }
111 }
112 }
113 }
```

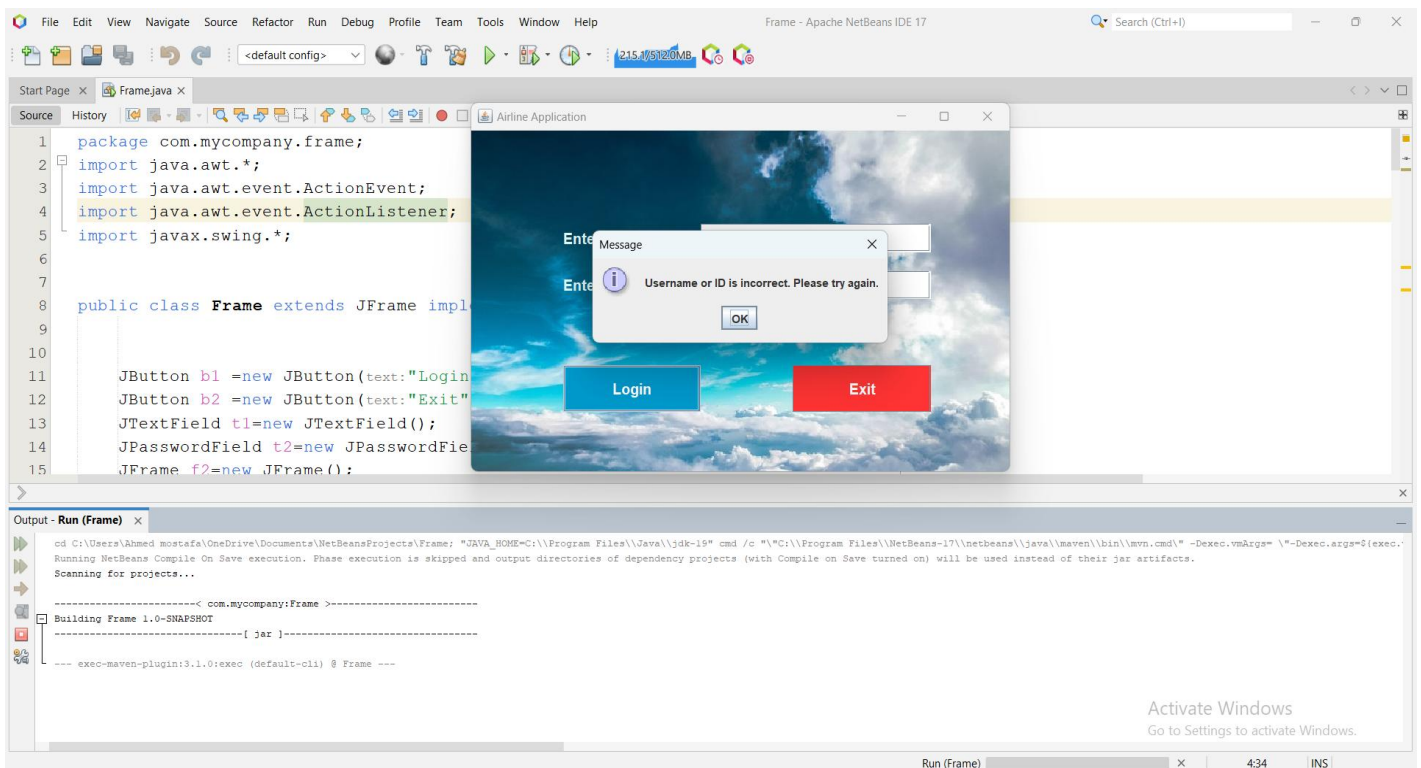
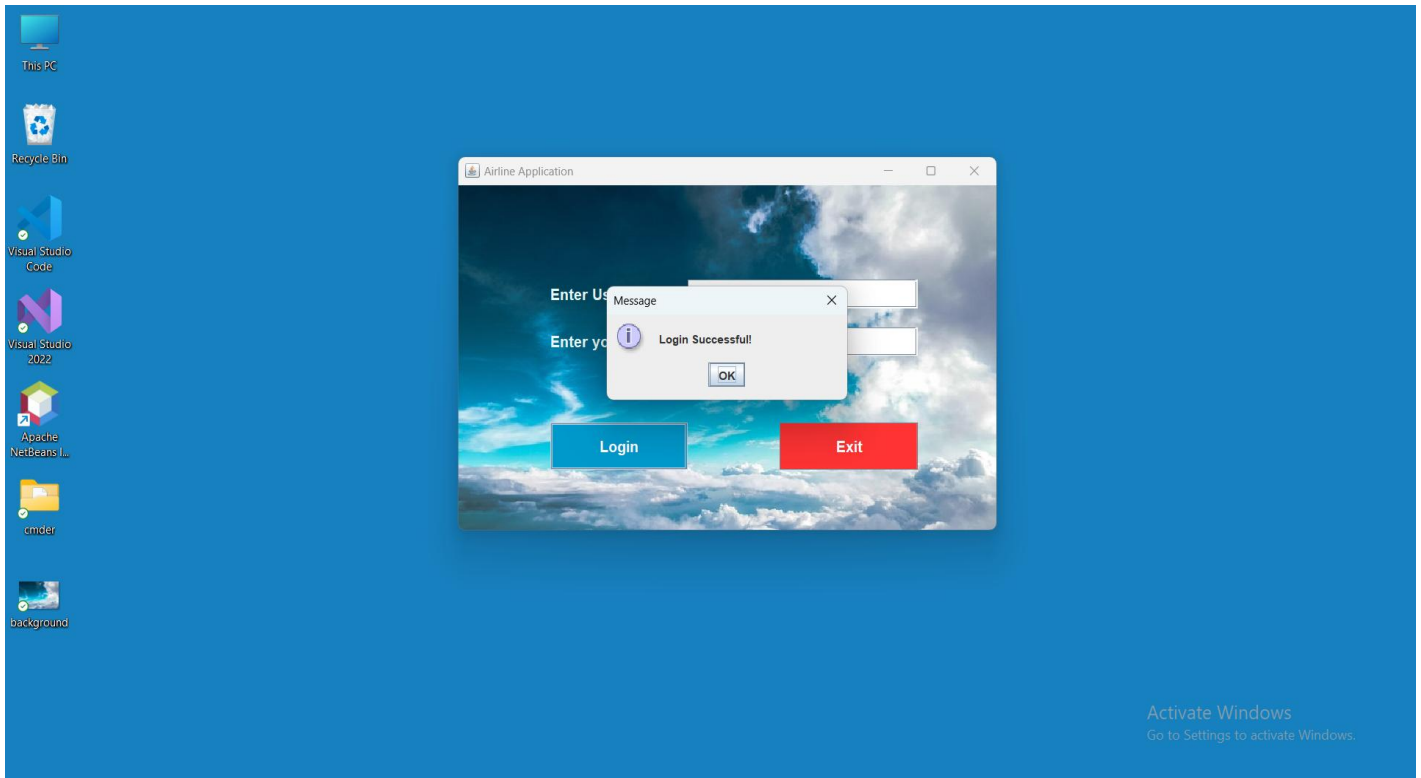
Activate Windows
Go to Settings to activate Windows.

com.mycompany.frame.Frame > Frame >

Run (Frame) X (14 more...) 71:15 INS Unix (LF)

➤ SHAPE OF THE FIST FRAME





➤ SHAPE OF THE SECOND AND THIRD FRAME WITH DRAG AND DROP

